Renjie LIU

Cambridge, MA, United States

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EDUCATION

Massachusetts Institute of Technology, *Cambridge, United States* **Sep. 2026 – Present**
Ph.D. in Computer Science — Advised by ***Prof. Samuel Madden*** *&* ***Prof. Michael Cafarella***

Southern University of Science and Technology, *ShenZhen, China* **Sep. 2023 – Jun. 2026**
M.phil. in Computer Science — Advised by ***Prof. Bo Tang***

Southern University of Science and Technology, *ShenZhen, China* **Sep. 2019 – Jun. 2023**
B.Eng. in Computer Science — Advised by ***Prof. Bo Tang*** *&* ***Prof. Xiao Yan***

RESEARCH INTERESTS

My research interest falls broadly in **systems for AI models and applications**. My past research focused on **systems for graph neural networks and retrieval**, spanning GPU kernels, distributed and out-of-core training recipes, and graph-based retrieval-augmented generation (GraphRAG). Currently, I work on **efficient systems for LLMs and agentic workloads**, such as attention mechanism, KV cache reuse and browser-use agents.

EXPERIENCE

NYU Systems Group, New York University **Dec. 2024 – Jun. 2025**
Student Intern — Advised by ***Prof. Jinyang Li***

- General and effective **GraphRAG** solutions and efficient **agentic applications**.

Shanghai AI Lab, Amazon Web Services **Jun. 2022 – Nov. 2024**
Applied Scientist Intern in ***DGL Team*** *— Advised by* ***Dr. Minjie Wang***

- Efficient **GNN training** on large-scale datasets in single-machine and distributed environments.

PUBLICATIONS

❋ **Accelerating Position-Independent Caching with Context-Adaptive KV Cache Compression**
Jie Xu, **Renjie Liu**, Yi Li, and Bo Tang
ACM Conference on Management of Data ***(SIGMOD 2027)***

❋ **FiX: Introducing Fine-grained Forget Gate into Softmax Attention**
Runzhong Li, **Renjie Liu**, Qing Li, and Bo Tang
International Conference on Machine Learning ***(ICML 2026)***

❋ **Efficient Neural-Symbolic Data System via Multi-Agent Collaboration**
Ye Yuan, Bo Tang, Zhaojing Luo, Boyang Li, **Renjie Liu**, and Zhilang Wei
IEEE International Conference on Data Engineering ***(ICDE 2026)***

❋ **PolyG: Adaptive Graph Traversal for Diverse GraphRAG Questions**
Renjie Liu, Haitian Jiang, Xiao Yan, Bo Tang, and Jinyang Li
https://arxiv.org/pdf/2504.02112 ***(2025)***

❋ **DiskGNN: Bridging I/O Efficiency and Model Accuracy for Out-of-Core GNN Training**
Renjie Liu*, Yichuan Wang*, Xiao Yan, Zhenkun Cai, Minjie Wang, Haitian Jiang, Bo Tang, and Jinyang Li
ACM Conference on Management of Data ***(SIGMOD 2025)***
*Equal Contribution

❋ **APT: Adaptive Parallel Training for Graph Neural Networks**
Kaihao Ma*, **Renjie Liu***, Zhenkun Cai, Xiao Yan, Xiang Song, Minjie Wang, and James Cheng
ACM Symposium on Principles and Practice of Parallel Programming ***(PPoPP 2025)***
*Equal Contribution

❋ **Forming Scalable, Convergent GNN Layers that Minimize a Sampling-Based Energy**
Haitian Jiang, **Renjie Liu**, Xiao Yan, Zengfeng Huang, Zhenkun Cai, Minjie Wang, and David Wipf
International Conference on Learning Representations ***(ICLR 2025)***

❋ **gSampler: General and Efficient GPU-based Graph Sampling for Graph Learning**
Ping Gong, **Renjie Liu**, Zunyao Mao, Zhenkun Cai, Xiao Yan, Cheng Li, Minjie Wang, and Zhuozhao Li
ACM Symposium on Operating Systems Principles ***(SOSP 2023)***

RESEARCH PROJECTS

Towards Efficient and Transparent Browser-use WebAgent **Mar. 2026 – Present**

✏ Design a new system abstraction for browser-use agents that modularizes execution into cooperative IR components and expresses actions as an explicit dataflow graph for efficient, transparent execution.

Fine-grained Forget Gate for Softmax Attention (FiX) **Oct. 2025 – Feb. 2026**

✏ Introduce a fine-grained forget gate into standard softmax attention, enabling the model to dynamically control what to retain and forget at a granular level. This is also the first V-O positional encoding.

Mixed-precision KV Cache Reuse for LLM Serving (CatKV) **Aug. 2025 – Jan. 2026**

✏ Build a position-independent KV caching system with context-adaptive mixed-precision compression, achieving 2–5× faster time-to-first-token and 6–10× higher compression ratios while preserving model accuracy.

Effective and Efficient Graph RAG Solution (PolyG) **Aug. 2024 – May 2025**

✏ Create the first Graph RAG benchmark covering diverse question types, and propose an adaptive approach that selects the best graph traversal strategy for each query pattern, achieving both effectiveness and efficiency.

Efficient out-of-core GNN Training on Large-scale Graphs (DiskGNN) **Sep. 2023 – July 2024**

✏ Build an out-of-core GNN training system for graphs exceeding CPU memory, using offline sampling with a four-level feature store and batched feature packing to achieve high I/O efficiency without hurting accuracy.

Adaptive Parallel Training for Graph Neural Networks (APT) **April 2023 – Dec. 2023**

✏ Build a general framework that automatically selects the best parallelism for distributed GNN training, using a unified matrix-based abstraction of parallel schemes and a cost model of the hardware and model.

General and Efficient GPU-based Graph Sampling (gSampler) **Aug. 2022 – April 2023**

✏ Build a high-performance GPU graph sampling framework for GNN training that expresses diverse sampling algorithms through matrix-centric APIs, with optimizations such as kernel fusion and format selection.

HONORS AND AWARDS

♣ China National Scholarship **Nov. 2024**
♣ University Merit Scholarship for Outstanding Student: The 1st Class **Sep. 2023, 2022**
♣ Outstanding Graduates of Department of Computer Science and Engineering **Jun. 2023**
♣ ACM SIGMOD 2022 Programming Contest: World's Finalist **April 2022**

TEACHING EXPERIENCE

❍ **TA: Computer Organization and Architecture** **Mar. 2022 – Jun. 2022**
❍ **TA: Introduction to Computer Programming** **Sep. 2020 – Jun. 2021**